

Interactive Climate Profile Concept

Hawaiian Coot Conservation Website

This document outlines the interactive “Climate Profile” section for the Hawaiian Coot endangered species website. The goal is to transform a static information section into an engaging educational experience that visually explains how Hawaii’s climate supports the Hawaiian Coot’s survival and habitat.

Main Objective

Create an immersive climate section that combines modern UI design, animation, and environmental storytelling. The interaction should help users understand the relationship between climate, wetlands, rainfall, temperature, and Hawaiian Coot survival.

Interactive Temperature Card

The temperature card represents Hawaii’s tropical climate.

Default State:

- Displays temperature range: 21°C → 32°C
- Warm red/orange glowing interface
- Thermometer icon centered at top

Hover Effects:

- Thermometer fills upward
- Heat-wave animations move across the card
- Number counter animates smoothly
- Glow intensity increases

Educational Message:

“Warm tropical temperatures allow wetlands and taro fields to remain active year-round, supporting feeding and nesting habitats.”

Interactive Rainfall Card

The rainfall card explains how precipitation sustains wetland ecosystems.

Default State:

- Displays annual rainfall: 500–2000 mm
- Blue glowing climate panel
- Rain cloud icon above the values

Hover Effects:

- Animated rain particles fall
- Water ripple effects appear
- Rainfall values animate into view
- Subtle ambient glow pulses

Educational Message:

“Heavy rainfall sustains freshwater ponds and marshes used by Hawaiian Coots for nesting and feeding.”

Advanced Interactive Ideas

1. Habitat Impact Meter

A small indicator dynamically shows how climate conditions affect:

- Wetland stability
- Food availability
- Nesting conditions

2. Climate Change Toggle

Users can compare:

- Current healthy climate conditions
- Reduced rainfall scenarios
- Wetland degradation
- Habitat loss

3. Island Climate Explorer

Hovering over islands like Oahu, Maui, or Kauai updates:

- Rainfall data
- Habitat descriptions
- Wetland health indicators

Why This Improves the TSA Project

The interactive climate section improves:

- Visual storytelling
- User engagement
- Educational value
- Scientific communication
- Modern web design quality

It also demonstrates:

- UI/UX thinking
- Interactive design
- Front-end animation concepts
- Environmental awareness

Recommended Technologies

React — Main framework

Framer Motion — Smooth animations and hover transitions

CSS Animations — Rain particles and glow effects

Intersection Observer — Scroll-triggered reveal animations

SVG Graphics — Lightweight animated climate icons

Conclusion

The Climate Profile section can become one of the most visually impressive and educational parts of the Hawaiian Coot website. By combining interactive storytelling with conservation education, the experience becomes memorable, informative, and highly engaging for TSA judges and viewers.